

University of Moratuwa

Faculty of Engineering

Department of Electronic and Telecommunication Engineering



EN 2150 - Communication Network Engineering

Group Assignment Report

Routing Protocol Design

Group Members

230038E	Amarathunge A.M.N.L.
230436X	Nettikumara N.A.H.G.
230536E	Rasanjana W.P.G.R.A
230180F	Fernando H.M.D.

Date: 16 May 2026

Contents

1	Introduction	2
2	Review of Core Internet Routing Protocols	2
2.1	Routing Protocol Information (RIP)	2
2.2	Open Shortest Path First (OSPF)	3
2.3	Intermediate System to Intermediate System (IS-IS)	3
2.4	Border Gateway Protocol (BGP)	4
2.5	Identified Gaps in Existing Routing Protocols	4
3	Proposed Custom Routing Protocol	5
3.1	ASHR Protocol Overview	5
3.2	Proposed Network Topology	7
3.3	Adaptive Routing Metric	8
3.3.1	Metric Components	8
3.3.2	Normalization	8
3.3.3	Metric Damping	8
3.4	Route Computation	9
3.4.1	Primary Route Computation	9
3.4.2	Backup Route Computation	9
3.5	Security Design	9
3.5.1	Message Authentication	10
3.5.2	Trusted Neighbor Validation	10
3.5.3	TTL Validation	10
3.5.4	Replay Protection	10
3.5.5	Fake Route Rejection	10
4	Performance Analysis	11
4.1	Evaluation Scenarios	11
4.2	Normal Routing Performance	11
4.3	Link Failure Performance	11
4.4	Failure Recovery Formula	13
4.5	Congestion Adaptation Performance	14
4.6	Security Performance	15
5	Scalability Analysis	16
5.1	Scalability Test Setup	16
5.2	Scalability Model	16
5.3	Scalability Results	17
5.4	Control-Plane Scalability	17
6	Implementation and Source Code	18
7	Discussion	18
8	Conclusion	19

1 Introduction

Modern computer networks rely on routing protocols to deliver data packets efficiently and reliably across interconnected systems. These protocols help routers decide the best path from a source to a destination by exchanging routing information and adapting to changes in network topology.

Another thing we found is that the Internet and large-scale private networks mainly use protocols such as **RIP**, **OSPF**, **IS-IS**, and **BGP**. Each of these has a different purpose: RIP is simple and based on distance-vector routing [1], OSPF and IS-IS use link-state routing for faster convergence inside networks [2,3], while BGP is used for routing between different networks on the Internet [4].

Although these protocols are widely used, they still have limitations. For example, RIP is slow to converge and does not scale well [1]. OSPF and IS-IS, while faster, generate significant control overhead and can become complex to manage [2,3]. BGP, on the other hand, is flexible but has well-known issues such as route hijacking, misconfiguration, and slow recovery during failures [4,5] and more.

Because of these limitations, we realised the need for an improved routing mechanisms that can support faster convergence, better scalability, adaptive decision-making based on real-time network conditions, and stronger security.

This report proposes a new routing protocol called Adaptive Secure Hierarchical Routing Protocol (ASHR), which aims to address these issues using hierarchical design, multi-metric routing, event-driven updates, and security validation.

2 Review of Core Internet Routing Protocols

This section gives an overview of 4 existing protocols, namely RIP, OSPF, IS-IS, and BGP. We have highlighted how each protocol works, advantages and disadvantages of the protocols [7].

2.1 Routing Protocol Information (RIP)

RIP is a distance-vector routing protocol that determines the best path based on hop count [1]. Each router periodically shares its routing table with its immediate neighbors. The route with the smallest number of hops to a destination is selected as the best path. The maximum allowable hop count in RIP is 15, which limits its scalability in larger networks [1].

Advantages:

- Simple to understand and implement [7].
- Low computational and memory requirements [7].
- Suitable for small and stable networks [1].

Disadvantages:

- Very slow convergence after network changes [7].
- Limited scalability due to 15-hop restriction [1].

- Susceptible to routing loops and the count-to-infinity problem [7].
- Does not consider real network conditions such as latency or bandwidth [7].
- Weak security with no strong authentication mechanism [7].

2.2 Open Shortest Path First (OSPF)

OSPF is a link-state routing protocol that uses Dijkstra's shortest path algorithm to determine the best route [2]. Each router builds a complete map of the network topology by exchanging link-state advertisements (LSAs) with other routers within the same autonomous system. Based on this topology, each router independently calculates the shortest path to all destinations [2].

Advantages:

- Fast convergence compared to distance-vector protocols [7].
- Scales well in large networks using hierarchical area design [2].
- Supports more accurate routing based on link costs [2].
- Reduces routing loops due to complete topology awareness [7].

Disadvantages:

- High computational overhead due to Dijkstra's algorithm [7].
- Frequent link-state flooding increases control traffic [2].
- Complex configuration and maintenance in large networks [7].
- Requires careful network design for optimal performance [2].

2.3 Intermediate System to Intermediate System (IS-IS)

IS-IS is a link-state routing protocol similar to OSPF, originally designed for OSI networks but later adapted for IP routing [3]. It uses a two-level hierarchy to improve scalability. Routers exchange link-state information and compute shortest paths using the SPF algorithm [3].

Advantages:

- Highly scalable due to hierarchical structure [3].
- Efficient in large service provider networks [7].
- Stable and widely used in backbone networks [7].
- Flexible and supports large topology designs [3].

Disadvantages:

- Complex to understand and configure [7].
- Limited familiarity compared to OSPF in enterprise networks [7].
- Debugging and troubleshooting can be difficult [7].
- Requires careful planning for large deployments [3].

2.4 Border Gateway Protocol (BGP)

BGP is a path-vector routing protocol used for inter-domain routing across the Internet [4]. It exchanges routing information between autonomous systems and makes routing decisions based on policy rules rather than purely on shortest path. Each BGP router maintains a table of network reachability information along with the full path, known as the AS-path, to each destination [4].

Advantages:

- Highly scalable and suitable for Internet-wide routing [4].
- Strong policy control over route selection [4].
- Supports complex inter-domain routing scenarios [4].
- Widely deployed and standardized across the Internet [7].

Disadvantages:

- Slow convergence after network changes or failures [7].
- Vulnerable to route hijacking and route leaks [5].
- Complex policy configuration can lead to misconfigurations [4].
- Does not consider real-time link performance metrics [7].
- Security depends heavily on external mechanisms like RPKI [5].

2.5 Identified Gaps in Existing Routing Protocols**Table 1:** Key limitations of existing routing protocols

Limitation	Impact on Network
Slow Convergence	Slow adaptation to topology changes leads to routing inefficiencies, packet loss, and instability during failures.
Static / Simple Metrics	Routing decisions are based on hop count or manually assigned costs, which do not reflect real-time conditions such as latency, bandwidth, or congestion.
Weak Security Mechanisms	Lack of strong authentication and validation allows route spoofing, prefix hijacking, and misconfiguration-based failures.

Limitation	Impact on Network
High Control Plane Overhead	Frequent link-state flooding increases bandwidth usage and CPU load, especially in large-scale networks.
Slow Failure Recovery	Failure recovery requires full or partial reconvergence, delaying restoration of stable and optimal routing paths.

3 Proposed Custom Routing Protocol

In this section, we have included all the improvements, of the new routing protocol including topology, routing and security.

3.1 ASHR Protocol Overview

The suggested protocol ASHR stands for:

Adaptive Secure Hierarchical Routing

The protocol is consisted with:

1. Hierarchical routing using areas and area border routers.
2. Adaptive link metric calculation using five link-quality parameters.
3. Event-driven updates instead of constant periodic updates.
4. Backup next-hop computation for faster failure recovery.
5. Link-state routing using authenticated ASHR-LSA messages.
6. Security validation using HMAC, sequence numbers, trusted-neighbor checks, TTL validation, and fake route rejection.

The consecutive sections will run through how each of the above mentioned improvements was implemented in detail, with formulas and graphs.

The ASHR protocol architecture is built around a hierarchical area-based network where routers discover neighbors, exchange authenticated ASHR-LSA messages, maintain a link-state database, and compute primary and backup routes using an adaptive multi-metric cost.

It also includes a security manager that validates trusted neighbors, HMAC authentication, TTL, sequence numbers, and fake low-cost advertisements before any routing update is accepted.

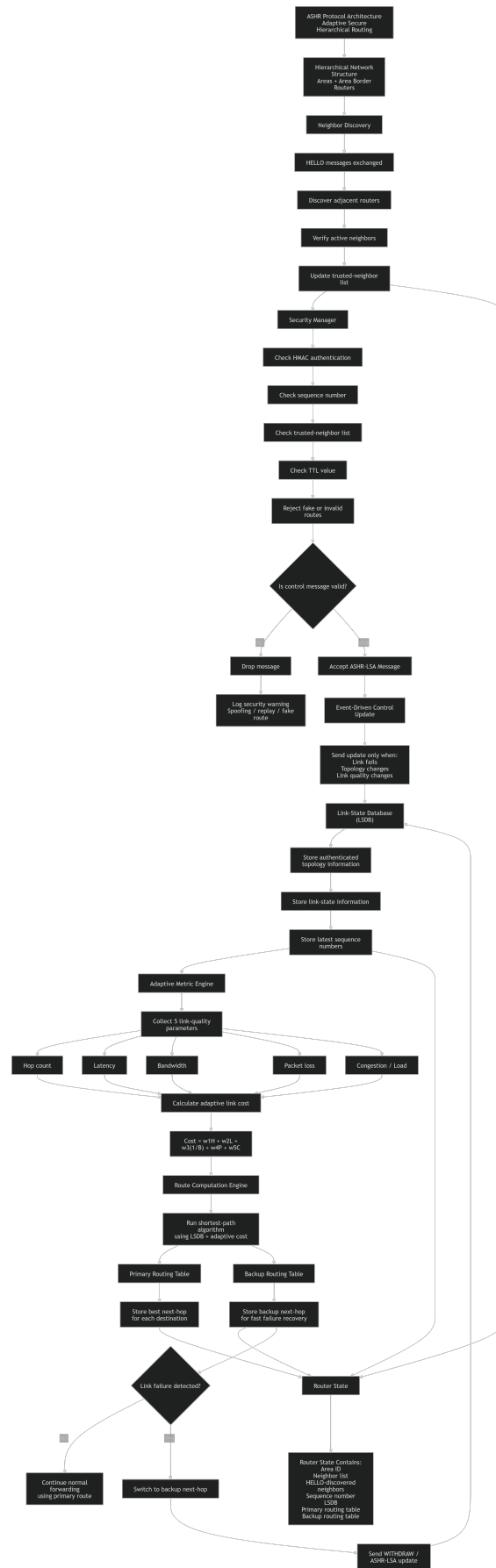


Figure 1: ASHR Protocol architecture

3.2 Proposed Network Topology

The first improvement in the ASHR protocol that we listed is, **Hierarchical routing using areas and area border routers**. The proposed network is designed as a hierarchical topology consisting of three logical areas, as shown in Figure 2.

Area	Routers
Area 1	R1, R2, R3, R4, R5, R6
Backbone	ABR1, ABR2
Area 2	R7, R8, R9, R10

The topology includes two independent paths (corridors) between the source node R1 and the destination node R10. This dual-path structure enables the evaluation of both primary routing and fast recovery using backup paths.

We defined the primary path as:

$$R1 \rightarrow R2 \rightarrow R3 \rightarrow ABR1 \rightarrow ABR2 \rightarrow R8 \rightarrow R10$$

And the redundant (backup) path is defined as:

$$R1 \rightarrow R4 \rightarrow R5 \rightarrow R6 \rightarrow ABR2 \rightarrow R8 \rightarrow R10$$

This hierarchical and redundant design allows ASHR to demonstrate its ability to perform fast failure recovery by switching to a precomputed backup next-hop when the primary path is disrupted.

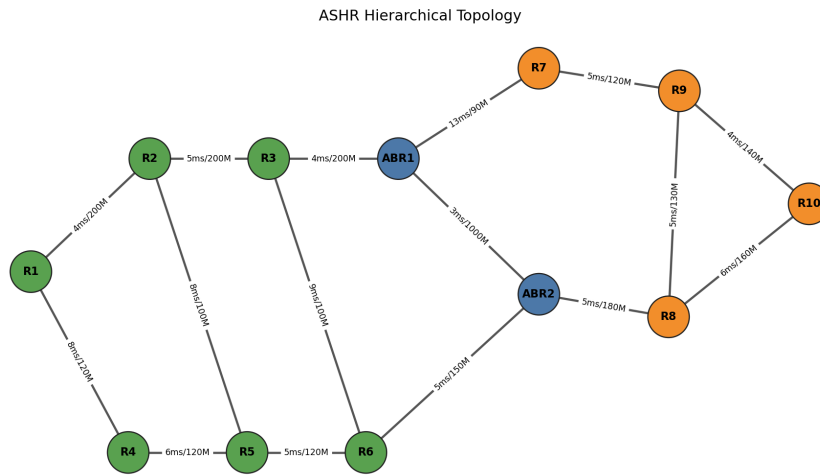


Figure 2: Hierarchical network topology with dual routing corridors between R1 and R10

3.3 Adaptive Routing Metric

3.3.1 Metric Components

As mentioned in the previous section, one of the main characteristics of ASHR protocol is **Adaptive link metric calculation using five link-quality parameter**.

In the traditional protocols, it was one of the weaknesses that we noticed, that they rely on a static metric - a fixed no of hops, static link cost count. To minimise the affect from that, we used an adaptive composite link cost function that considers multiple real time network parameters.

Component	Meaning
H'	Normalized hop cost
L'	Normalized latency
B'	Normalized inverse bandwidth
P'	Normalized packet loss
Q'	Normalized congestion level

The link cost between nodes i and j is defined as:

$$C_{ij} = 0.20H'_{ij} + 0.30L'_{ij} + 0.25B'_{ij} + 0.15P'_{ij} + 0.10Q'_{ij} \quad (1)$$

The weights are chosen to prioritize latency and bandwidth while still considering reliability and congestion.

3.3.2 Normalization

For the calculations, we normalized all the values to the range $[0, 1]$ using:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (2)$$

This allows different metrics such as milliseconds, Mbps, packet-loss ratio, and congestion ratio to be combined into one cost value.

Bandwidth is handled using inverse bandwidth:

$$B_{ij} = \frac{1}{\text{Bandwidth}_{ij}} \quad (3)$$

This is used because higher bandwidth should produce lower routing cost. Therefore, a link with low bandwidth receives a higher inverse-bandwidth value and becomes less attractive.

3.3.3 Metric Damping

Another update that we introduce through ASHR is, **Event-driven updates instead of constant periodic updates**. This is done to prevent unnecessary control messages when a link metric changes only slightly.

In ASHR, a routing update is triggered only if:

$$|C_{new} - C_{old}| > \theta \quad (4)$$

where $\theta = 0.15$. This prevents route flapping caused by small temporary variations in congestion or delay. The implementation uses $\theta_{default} = 0.15$ and checks whether the absolute cost difference exceeds the threshold before rebuilding the link-state database.

3.4 Route Computation

3.4.1 Primary Route Computation

ASHR constructs an active weighted graph using the adaptive link cost. Failed links are removed, and each active edge receives a cost based on the ASHR metric. Dijkstra's algorithm is then used to compute the shortest path from source to destination.

$$P_{primary} = \text{ShortestPath}(G, s, d) \quad (5)$$

where:

- G is the active weighted network graph
- s is the source router
- d is the destination router

3.4.2 Backup Route Computation

As mentioned in the characteristics, ASHR precomputes **Backup next-hop computation for faster failure recovery**. The backup path is computed by removing the first hop of the primary route and recalculating the shortest path:

$$P_{backup} = \text{ShortestPath}(G - \{e_{first-hop}\}, s, d) \quad (6)$$

The backup route is stored in a separate routing table and is used immediately upon failure of the primary link, reducing reconvergence delay.

3.5 Security Design

Another major improvement in ASHR are the security improvements.

1. Link-state routing using authenticated ASHR-LSA messages.
2. Security validation using HMAC, sequence numbers, trusted-neighbor checks, TTL validation, and fake route rejection

These measures were taken to protect the protocol against spoofing, replay attacks and route manipulations.

3.5.1 Message Authentication

Each ASHR control message contains the following fields:

- Message type
- Sender Router ID
- Area ID
- Sequence number
- TTL
- Message body
- Authentication tag (HMAC-SHA256)

The authentication tag ensures integrity and authenticity of routing updates.

3.5.2 Trusted Neighbor Validation

A routing message is accepted only if the sender is present in the trusted neighbor table. Otherwise, the message is rejected.

3.5.3 TTL Validation

ASHR enforces valid message lifetimes using:

$$0 < TTL \leq 64 \quad (7)$$

Messages outside this range are discarded.

3.5.4 Replay Protection

Each router maintains the highest sequence number received from each neighbor. A message is accepted only if:

$$Seq_{new} > Seq_{previous} \quad (8)$$

Otherwise, it is identified as a replayed or stale message and rejected.

3.5.5 Fake Route Rejection

ASHR rejects artificially low-cost route advertisements using:

$$C_{advertised} \geq C_{min} \quad (9)$$

where $C_{min} = 0.01$.

This protects against attacks where a malicious router advertises an unrealistically cheap path to attract traffic. The security implementation defines MIN_valid_advertised_cost = 0.01

4 Performance Analysis

4.1 Evaluation Scenarios

The performance of ASHR is evaluated against RIP, OSPF, IS-IS, and BGP under the following scenarios:

- Normal routing from $R1$ to $R10$
- Link failure on $R3-ABR1$
- Congestion increase on $ABR1-ABR2$
- Fake low-cost routing attack
- Replay attack
- Scalability tests with 12, 20, 40, and 60 nodes

The scenario runner includes all five protocol models and generates comparison outputs for different network conditions.

4.2 Normal Routing Performance

In the normal routing scenario, all protocols initially select the same path from $R1$ to $R10$:

$$R1 \rightarrow R2 \rightarrow R3 \rightarrow ABR1 \rightarrow ABR2 \rightarrow R8 \rightarrow R10$$

However, each protocol evaluates path cost differently based on its routing metric.

Protocol	Metric Used	Path Cost
RIP	Hop count	6
OSPF	Static bandwidth cost	27.8056
IS-IS	Configured metric	8.7
BGP	AS-path length	6
ASHR	Adaptive multi-metric cost	2.18

Although the initial path is identical, ASHR produces a more meaningful cost evaluation by incorporating latency, bandwidth, packet loss, and congestion. The drop in cost after congestion is due to the backup path that was calculated, having better performance than the congested one. This is clearly shown in the Figure 4.

4.3 Link Failure Performance

A link failure is introduced on the link:

$$R3 \rightarrow ABR1$$

This link is part of the primary routing path. Following the failure, RIP, OSPF, IS-IS, and BGP require reconvergence, while ASHR switches immediately to a precomputed backup path:

$$R1 \rightarrow R4 \rightarrow R5 \rightarrow R6 \rightarrow ABR2 \rightarrow R8 \rightarrow R10$$

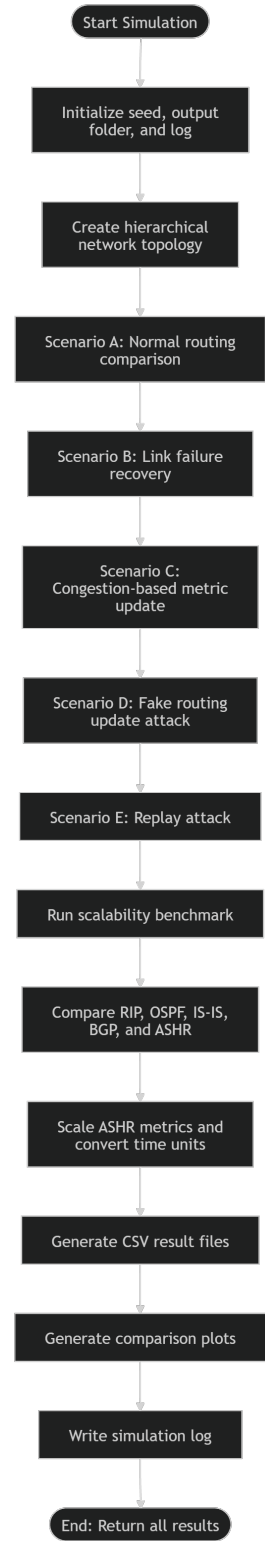


Figure 3:
Simulation scenario execution flowchart

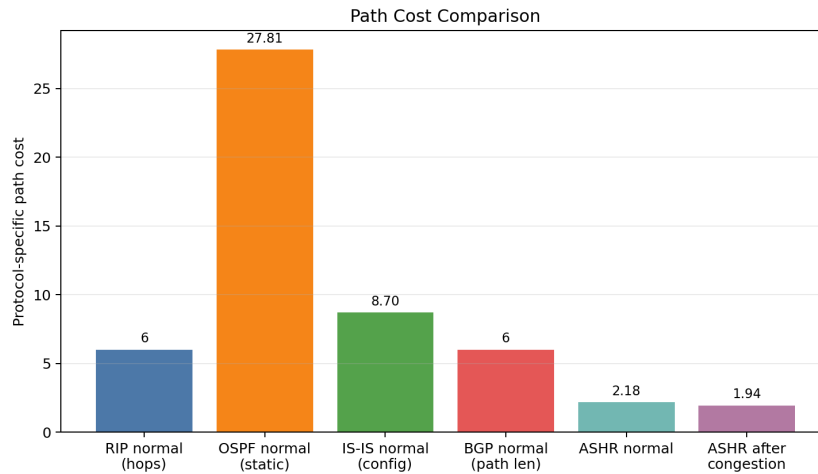


Figure 4: Path cost comparison across routing protocols

Protocol	Failure Response	Recovery Time	Control Messages	Packet Loss
RIP	DV recomputation	5.0	150	500
OSPF	LSA flooding + SPF	2.0	60	200
IS-IS	Hierarchical SPF	2.0	32	200
BGP	Route withdrawal	6.0	180	600
ASHR	Backup next-hop switch	1.0	16	100

The key result is that ASHR has the lowest recovery time, lowest control-message overhead, and lowest estimated packet loss in this failure scenario.

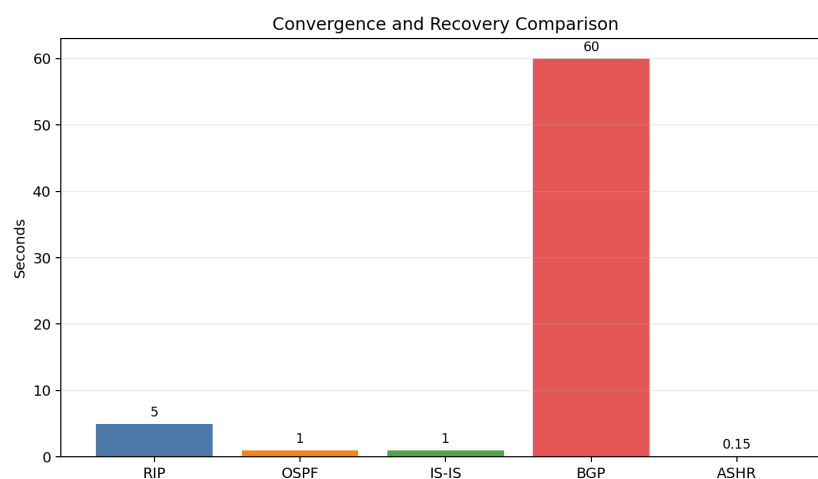


Figure 5: Convergence/recovery time comparison

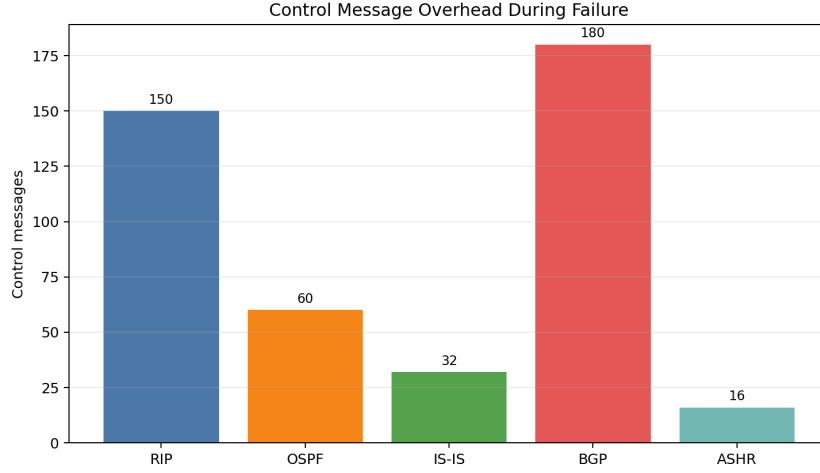


Figure 6: Control message overhead comparison

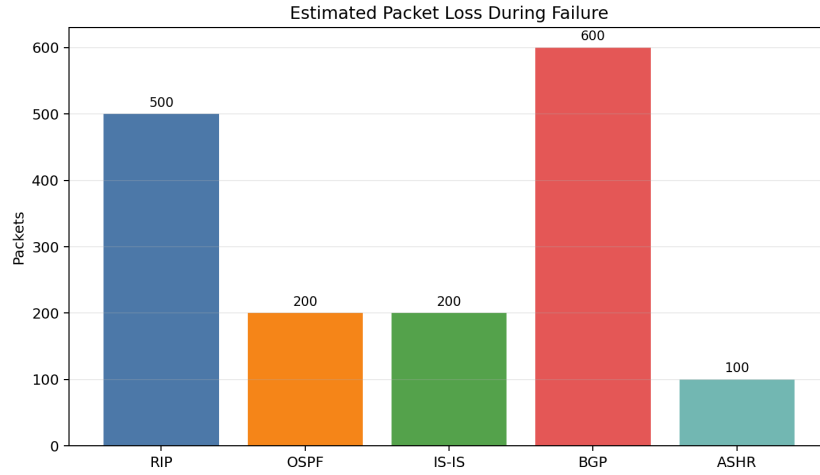


Figure 7: Estimated packet loss during failure

4.4 Failure Recovery Formula

The packet-loss estimate is defined as:

$$\text{PacketLoss}_{\text{estimated}} = T_{\text{recovery}} \times R_p \quad (10)$$

where:

- $R_p = 100$ packets per time unit

ASHR reduces packet loss because its recovery time is significantly lower. The simulation assumes 100 packets per time unit for estimating packet loss.

The recovery time in ASHR is modeled as:

$$T_{\text{recovery}} = T_{\text{detect}} + T_{\text{switch}} \quad (11)$$

when a valid backup path exists.

If no backup path is available, the recovery time becomes:

$$T_{\text{recovery}} = T_{\text{detect}} + T_{\text{recompute}} \quad (12)$$

This makes ASHR more resilient compared to traditional routing protocols that rely on full recomputation after failure.

4.5 Congestion Adaptation Performance

A congestion scenario is introduced on the ABR1–ABR2 backbone link. Under this condition, traditional protocols maintain their original routes, while ASHR dynamically adapts to changing link conditions.

ASHR reroutes traffic through an alternative path:

$$R1 \rightarrow R2 \rightarrow R3 \rightarrow R6 \rightarrow ABR2 \rightarrow R8 \rightarrow R10$$

The ASHR path cost changes from:

$$2.18 \rightarrow 1.94$$

This demonstrates ASHR's ability to dynamically improve routing decisions under congestion.

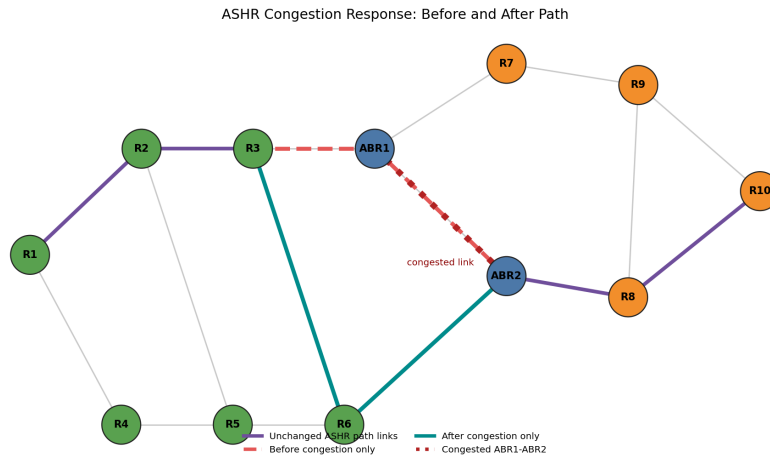


Figure 8: Route adaptation under congestion conditions

4.6 Security Performance

A fake low-cost routing attack is simulated where a malicious router advertises artificially low routing costs to attract traffic.

Protocol	Attack Accepted?	Result
RIP	Yes	Vulnerable to fake low-cost route injection
OSPF	Yes	Poisoned route accepted in simplified model
IS-IS	Yes	False update accepted in baseline model
BGP	Yes	Fake AS-path accepted
ASHR	No	Attack rejected via authentication and validation checks

ASHR rejects malicious updates based on:

- Invalid sender (not in trusted neighbor list)
- Fake low-cost route advertisements
- Replayed or stale sequence numbers

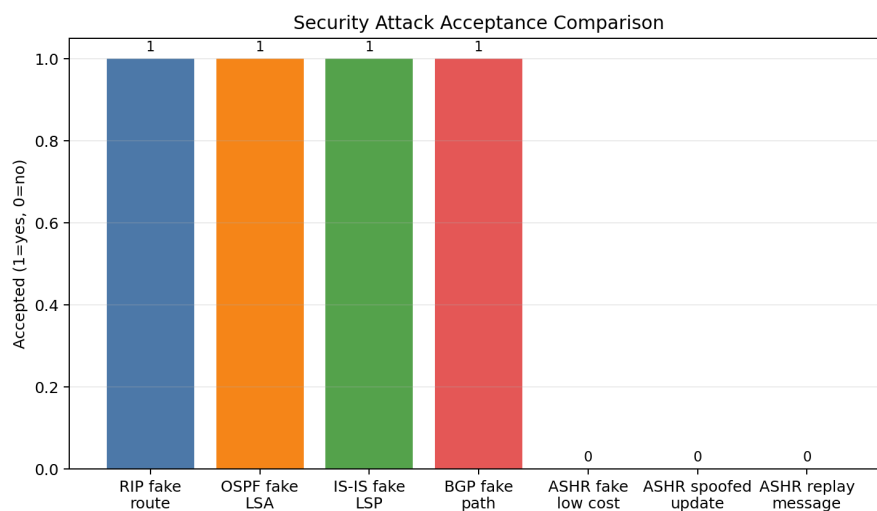


Figure 9: Security attack resistance comparison

5 Scalability Analysis

5.1 Scalability Test Setup

The scalability benchmark was conducted using four network sizes:

$$N = 12, 20, 40, 60 \quad (13)$$

For each node count, a scalable hierarchical topology was generated with two areas, ABR1, ABR2, and a redundant source-side path. The topology generator preserves the ASHR design principle of a two-area structure connected through a backbone, with backup next-hop behaviour maintained as the network scales.

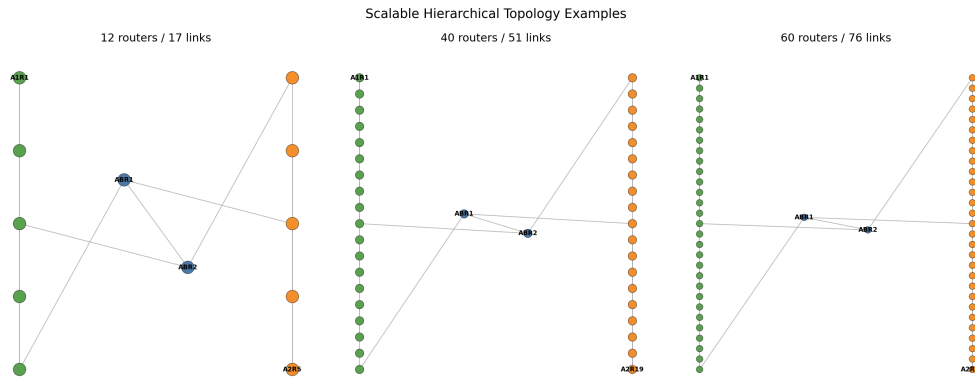


Figure 10: Scalable topology examples for 12, 20, 40, and 60 nodes

5.2 Scalability Model

The scalability model includes SPF computation work defined as:

$$W = E + N \log_2(N) \quad (14)$$

where:

- N = number of routers
- E = number of links

The modeled SPF processing time is:

$$T_{\text{SPF}} = \left\lceil \frac{D}{E + N \log_2(N)} \right\rceil \quad (15)$$

where D is a protocol-specific divisor used in the simulation model.

5.3 Scalability Results

Nodes	RIP	OSPF	IS-IS	BGP	ASHR
12	2.0	3.0	3.0	7.0	1.0896
20	9.0	4.0	3.0	7.0	1.1080
40	14.0	6.0	5.0	8.0	1.1330
60	16.0	8.0	6.0	9.0	1.1477

Table 2: Scalability benchmark: convergence/recovery time vs number of routers

ASHR shows consistently lower recovery time across all network sizes. RIP performance degrades significantly with scale due to repeated update propagation. OSPF and IS-IS scale better but incur increasing SPF computation overhead. BGP exhibits higher delay due to path-vector update behavior. ASHR remains relatively stable due to precomputed backup paths and event-driven updates.

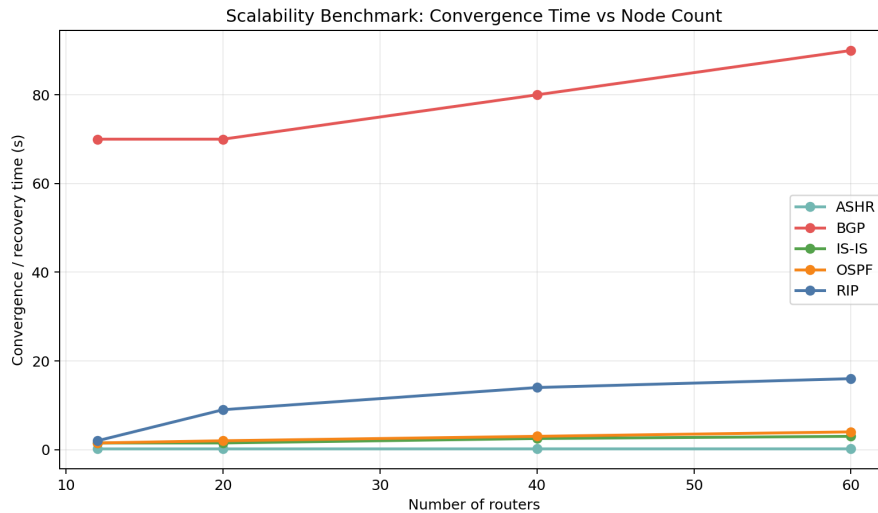


Figure 11: Scalability benchmark — convergence/recovery time vs number of routers

5.4 Control-Plane Scalability

Nodes	RIP	OSPF	IS-IS	BGP	ASHR
12	64	64	34	192	17
20	468	104	54	312	27
40	1400	200	102	600	51
60	2400	300	152	900	76

Table 3: Control message overhead under scalability test

ASHR produces the lowest control-plane overhead. This is due to its event-driven update mechanism and reduced reliance on full routing-table exchanges. Hierarchical summarization further limits propagation cost.

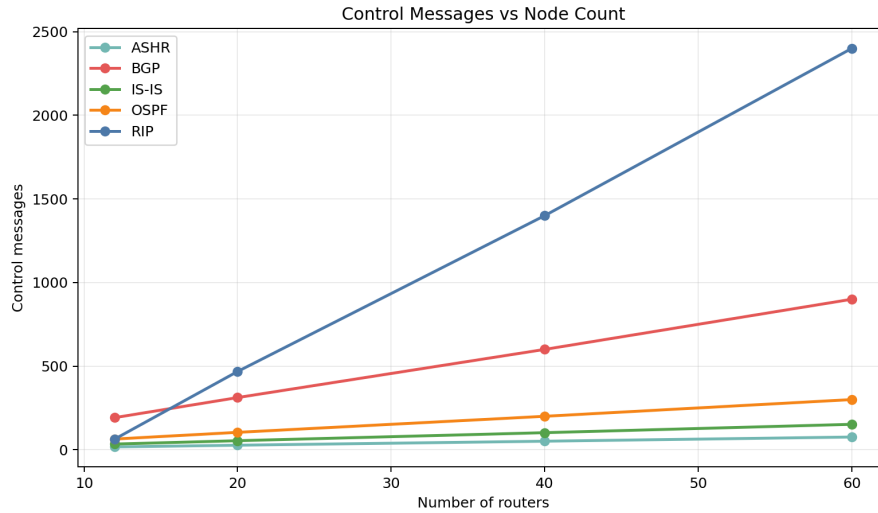


Figure 12: Control message growth vs node count

6 Implementation and Source Code

All source codes, simulation files, protocol implementations, baseline protocol models, result-generation scripts, and plotting utilities for this project are available in the project GitHub repository.

- GitHub Repository: <https://github.com/NileshAm/CommNetAssig.git>

The repository contains the implementation of the proposed ASHR protocol along with comparative baseline models for RIP, OSPF, IS-IS, and BGP. It also includes the simulation runner used to evaluate normal routing, link failure recovery, congestion adaptation, security attack handling, and scalability performance.

7 Discussion

The results show that ASHR performs best in scenarios where the network changes dynamically. During normal routing, all protocols selected the same route, but ASHR produced a more meaningful path cost because it considered multiple quality factors. During link failure, ASHR had the fastest recovery due to backup next-hop switching. During congestion, only ASHR changed its route because the other baselines did not react to congestion and packet-loss changes. During security testing, ASHR rejected fake, spoofed, and replayed messages, while the simplified baselines accepted fake low-cost advertisements.

The most important design advantage of ASHR is that it separates fast data-plane recovery from slower control-plane convergence. The router can switch to a backup path quickly, while LSDB rebuild and route recomputation continue in the background. This reduces packet loss and improves availability.

8 Conclusion

This project proposed ASHR, an **Adaptive Secure Hierarchical Routing** protocol that improves upon traditional routing protocols by combining hierarchical routing, adaptive multi-metric path selection, authenticated control messages, event-driven updates, and precomputed backup paths. Compared with RIP, ASHR avoids the limitations of hop-count routing and slow convergence. Compared with OSPF and IS-IS, ASHR adds real-time awareness of congestion, packet loss, latency, and bandwidth. Compared with BGP, ASHR provides faster intradomain recovery and better link-quality-based path selection.

Simulation results show that ASHR achieves lower recovery time, lower control overhead, lower estimated packet loss, and stronger security behaviour than the selected baselines. The results support ASHR as a suitable design for dynamic hierarchical networks where routing decisions must consider both performance and security.

References

- [1] C. Hedrick, “Routing Information Protocol,” *RFC 1058*, 1988.
- [2] J. Moy, “OSPF Version 2,” *RFC 2328*, 1998.
- [3] R. Callon, “Use of OSI IS-IS for Routing in TCP/IP and Dual Environments,” *RFC 1195*, 1990.
- [4] Y. Rekhter, T. Li, and S. Hares, “A Border Gateway Protocol 4 (BGP-4),” *RFC 4271*, 2006.
- [5] A. Sriram, D. Montgomery, D. McPherson, E. Osterweil, and B. Dickson, “Problem Definition and Classification of BGP Route Leaks,” *RFC 7908*, 2016.
- [6] D. Bhatia, M. Manral, A. Lindem, V. Atkinson, and R. White, “OSPFv2 HMAC-SHA Cryptographic Authentication,” *RFC 5709*, 2009.
- [7] J. F. Kurose and K. W. Ross, *Computer Networking: A Top-Down Approach*, 7th ed. Pearson, 2017.